

Directional Derivatives and the Gradient

1. Rates of change in any direction

The partial derivatives f_x and f_y measure how fast $f(x, y)$ changes when we move parallel to the x - or y -axis. To measure the rate of change in an arbitrary direction we use a **unit vector** $u = \langle a, b \rangle$ with $|u| = 1$.

2. Definition

The directional derivative of f at (x_0, y_0) in the direction of u is

$$D_u f(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}$$

provided the limit exists.

THEOREM

If f is differentiable, then for every unit vector u :

$$D_u f = \nabla f \cdot u$$

where $\nabla f = \langle f_x, f_y \rangle$ is the gradient of f .

3. Example

Let $f(x, y) = x^2y + y^3$. Find the rate of change of f at $(1, 2)$ in the direction of $v = \langle 3, 4 \rangle$.

Normalize first: $|v| = 5$, so $u = \langle \frac{3}{5}, \frac{4}{5} \rangle$.

Gradient: $\nabla f = \langle 2xy, x^2 + 3y^2 \rangle$, so $\nabla f(1, 2) = \langle 4, 13 \rangle$.

Then $D_u f(1, 2) = 4 \cdot \frac{3}{5} + 13 \cdot \frac{4}{5} = \frac{64}{5}$.

4. Remarks

- Always use a unit vector. Using v instead of u would scale the answer by $|v|$.
- $D_u f$ is largest when u points in the direction of ∇f , and the maximum value is $|\nabla f|$.
- The gradient is perpendicular to the level curve $f(x, y) = c$ through the point.